

MASTER UPDATE PROMPT – HOSPITAL MANAGEMENT SYSTEM (NEPAL)

Project Overview

Update my existing Django-based Hospital Management System. Do **NOT** rebuild it from scratch. Analyze the existing project structure, models, views, URLs, templates, permissions, workflow, and UI before making changes.

Reuse the existing architecture and maintain consistency with the project.

The system should be professional, secure, scalable, and easy to use.

The project is developed for hospitals in Nepal.

IMPORTANT

Before modifying anything:

- Analyze the complete project.
 - Analyze the database models.
 - Analyze all modules.
 - Reuse existing code wherever possible.
 - Do not duplicate functionality.
 - Keep the UI consistent.
 - Improve the workflow instead of replacing it.
 - Preserve existing data relationships.
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TECHNOLOGY

- Django
- Django Templates
- Bootstrap
- JavaScript
- PostgreSQL/SQLite (existing database)
- Jasmine Django Admin theme (django-jazzmin)
- Existing GitHub template should be reused.

Do not replace the design.

Improve it.

SYSTEM MODULES

The system contains these modules:

- Super Admin
 - Registration Counter
 - Doctors
 - Laboratory
 - Radiology
 - Pharmacy
 - Nursing
 - Admission
 - Operation Theater
 - Blood Bank
 - Insurance
 - Medical Records
 - Accounts
 - Cash Counter
 - Patient Portal
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ROLE BASED ACCESS CONTROL

Implement strict Role-Based Access Control (RBAC).

Super Admin

Full control over everything.

Permissions:

- View everything
- Add
- Edit
- Delete
- Restore
- Print
- Export
- Reports
- User Management

- Role Management
- Financial Reports
- Dashboard
- Django Admin Access

Only Super Admin can:

- Delete records permanently
- Edit old records
- Print Patient ID Card
- Manage users
- Access Django Admin

Registration Counter

Permissions:

- Register patient
- Generate Hospital ID
- Generate OPD Ticket
- Search patient
- View patient history
- View medical records (Read Only)
- Print OPD Ticket
- Edit registrations within 24 hours only

Cannot:

- Delete permanently
- Edit after 24 hours

After 24 hours:

Only Super Admin can modify records.

Registration dashboard should show:

Today's:

- Total Patients
- Total Collection

Monthly

Yearly

Custom Date Range

Display values like:

Instead of

125000

show

1.25 Lakh

Clicking the box opens a popup modal with detailed information.

Doctors

Doctors can:

Search Patient

View:

- History
- Lab Reports
- Radiology
- Pharmacy
- Admission
- Nursing
- Blood Bank
- Medical Records

Doctor can:

- Write Notes
- Diagnosis
- Prescription
- Admit Patient
- Request Laboratory
- Request Radiology
- Refer Operation Theater
- Refer Blood Bank

Doctor cannot delete records.

Doctor Dashboard

Show:

Today's Patients

Monthly Patients

Yearly Patients

Custom Date

Graphs

Statistics

Doctor Quota System

Every doctor can define OPD quota.

Example:

Urology

Maximum:

100 Patients

Time:

9 AM – 2 PM

When quota is full:

Registration Counter cannot issue more tickets.

Online Registration also follows quota.

Laboratory

Can:

Search patient

Upload:

PDF

Scanned Reports

Automatically attach reports to Medical Record.

View only Laboratory reports.

Cannot edit reports after final submission.

Radiology

Same workflow as Laboratory.

Supports:

- PDF
- JPG
- Scan Upload

Automatically linked with Medical Record.

Pharmacy

Features:

Patient Search

Billing

Medicine Sales

Statistics

Filter by

Patient

Month

Year

Date

View:

Total Sales

Daily

Monthly

Yearly

Nursing

Can:

Search Patient

View History

View Medical Record

Cannot edit.

Admission

Can:

Search Patient

Admit Patient

Bed Management

Ward Management

Medical Record View

Admission Statistics

Operation Theater

Can access:

Everything related to surgery.

Can view:

Patient History

Lab

Radiology

Medical Record

Doctor Notes

Operation Notes

Blood Bank

Can:

View patient

Blood Requests

Blood History

Own Blood Records

Cannot modify other departments.

Insurance

Nepal Government Insurance Prototype

Support:

QR Scan

Insurance Number Search

Government Insurance

Private Insurance

Workflow:

Scan QR

OR

Enter Insurance Number

Retrieve Insurance

Approve Claim

Calculate Discount

Generate Insurance Receipt

Receipt contains:

Patient Information

Insurance Number

Insurance QR

Patient QR

Services

Discount

Amount Paid

Remaining Balance

Can export:

PDF

JPG

Automatically attach receipt to Medical Record.

Patient can view Insurance History.

Medical Records

Centralized repository.

Departments automatically upload reports.

View Access:

Doctor

Registration

Admission

Nursing

OT

Blood Bank

Super Admin

Medical Record Officer

Editing:

Only Admin.

Others:

Read Only.

Download:

PDF

Print

Accounts

Dashboard

Statistics

Daily Collection

Monthly

Yearly

Custom Range

Department-wise Collection

Registration

Laboratory

Radiology

Pharmacy

Admission

Insurance

Cash Counter

Drill-down reports.

Click Registration Collection

↓

Shows all patients

↓

Individual bills

↓

Payment history

Cash Counter

Handles:

OPD

Laboratory

Radiology

ECG

Ultrasound

X-Ray

All Services

Billing Interface:

Search Patient

Add Services

Autocomplete:

Typing

X

↓

Shows

X-Ray

Typing

E

↓

ECG

Typing

U

↓

Ultrasound

Fast searchable dropdown.

Payment Methods

Default:

Cash

Secondary:

eSewa

Support discounts:

Staff

Insurance

Special Discount

Receipt:

Print

PDF

JPG

PATIENT PORTAL

Separate Login.

Not inside Dashboard.

Professional website.

Patient Registration

Fields:

First Name

Last Name

Gender

Phone

Email

Password

Confirm Password

Phone Verification

Hospital ID generated automatically.

Hospital IDs should remain sequential regardless of whether registration occurs online or at the registration counter.

Example:

Online

H-000001



Counter

H-000002



Online

H-000003

Shared sequence.

Patient Login

Login by:

Hospital ID

or

Phone Number

Password

Forgot Password

Recovery only through phone number (OTP).

Patient Dashboard

Profile

Medical Records

Lab Reports

Radiology

Doctor Notes

Insurance

Appointments

Billing History

Download:

PDF

Cannot print Patient ID Card.

Only Registration Counter and Super Admin can print Patient ID Cards.

PATIENT REGISTRATION

Compulsory Fields

First Name

Last Name

Gender

Phone

Address

Age

Age should support:

Years

Months

Weeks

Days

Example:

24 Years

10 Months

15 Days

Registration Fees

New Patient

NPR 100

Old Patient

NPR 50

Configurable.

SEARCH SYSTEM

Every module should support searching by:

Hospital ID

Phone Number

QR Code

Patient Name

Insurance Number

QR Code

Every patient has QR Code.

Scanning opens patient profile instantly.

MEDICAL RECORD AUTOMATION

Automatically attach:

Lab Reports

Radiology

Insurance Claims

Doctor Notes

Operation Notes

Discharge Summary

Everything stored chronologically.

DASHBOARD

Every department should display:

Today's Work

Monthly Work

Yearly Work

Total Collection

Statistics

Charts

Recent Activities

Clickable statistic cards.

Large values should display in short form:

100000

↓

1 Lakh

2500000

↓

25 Lakh

Click opens popup modal with complete details.

USER INTERFACE

Professional hospital design.

Responsive.

Modern.

Simple.

Fast.

Every page should include:

Back Button

Breadcrumb

Popup Modals

Search Bar

Filter

Export

Print

Consistent buttons.

DJANGO ADMIN

Only Super Admin.

Dashboard contains:

"Django Admin"

Click opens Django Admin.

Use:

django-jazzmin

Configure:

Logo

Icons

Sidebar

Colors

Professional admin interface.

SECURITY

Implement:

Role Permissions

Audit Logs

CSRF

Authentication

Session Security

Password Hashing

Input Validation

Secure File Uploads

Protected Downloads

No unauthorized access.

REPORT EXPORTS

Support:

PDF

JPG

Printing

BUSINESS RULES

- Registration can edit records only within 24 hours.
 - Super Admin can edit anytime.
 - Patient ID Cards can only be printed by Registration Counter or Super Admin.
 - Patients cannot access other patients' records.
 - Medical Records are read-only for most departments.
 - Insurance claims automatically calculate discounts.
 - Doctor quotas limit appointment booking.
 - Online registration and counter registration share the same patient numbering sequence.
 - QR codes identify patients across all departments.
 - Default payment method is Cash, with eSewa available as an alternative.
 - All uploaded reports automatically become part of the patient's Medical Record.
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FINAL REQUIREMENTS

Before finishing the update:

- Review every module for consistency.
- Ensure all permissions work correctly.
- Verify all workflows end-to-end.
- Test role-based access.
- Test patient registration.
- Test billing.
- Test insurance.
- Test medical record uploads.
- Test patient portal.
- Ensure dashboards are responsive.
- Ensure reports and statistics are accurate.
- Maintain clean, modular, well-documented code.
- Do not remove existing features unless they are replaced with an improved implementation.
- Reuse the existing project structure and GitHub template while enhancing usability, security, and maintainability.